



Jenkins as a Service (JaaS)

A Road to manage your finances better



Rohit Agarwal
Product Owner

Experienced IT program manager specializing in infrastructure, security, and SRE management.



Meng Ni
Scrum Master

Skilled in agile coordination and platform efficiency improvement.



Kush Shah
Quality Assurance/Tester

Experienced in software testing and risk management for secure IT systems.



Mutian He
Software Developer

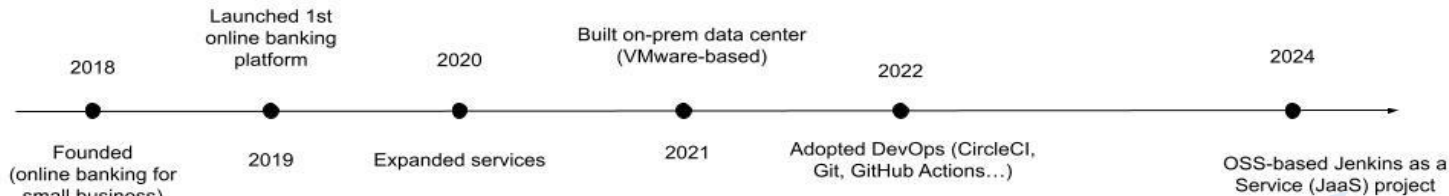
Focused on developing and integrating applications in cloud and PaaS environments.



Het Bhavesh Shah
Solution Architect

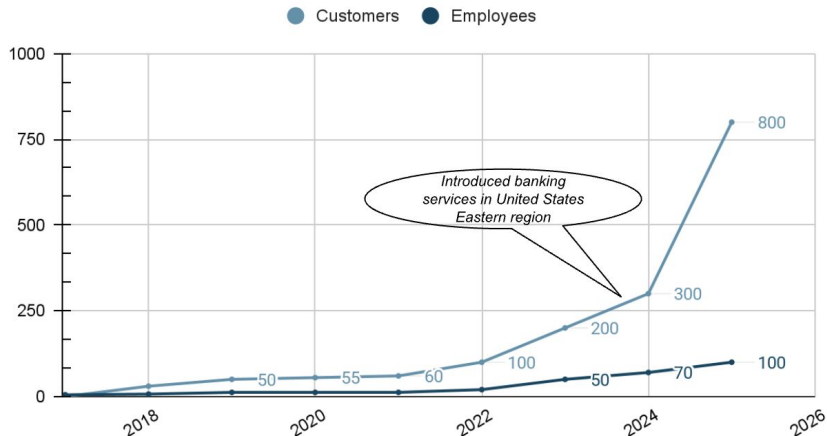
Expert in PaaS integration, enterprise security, and resilient cloud design.

PROJECT SCOPE AND CONTEXT



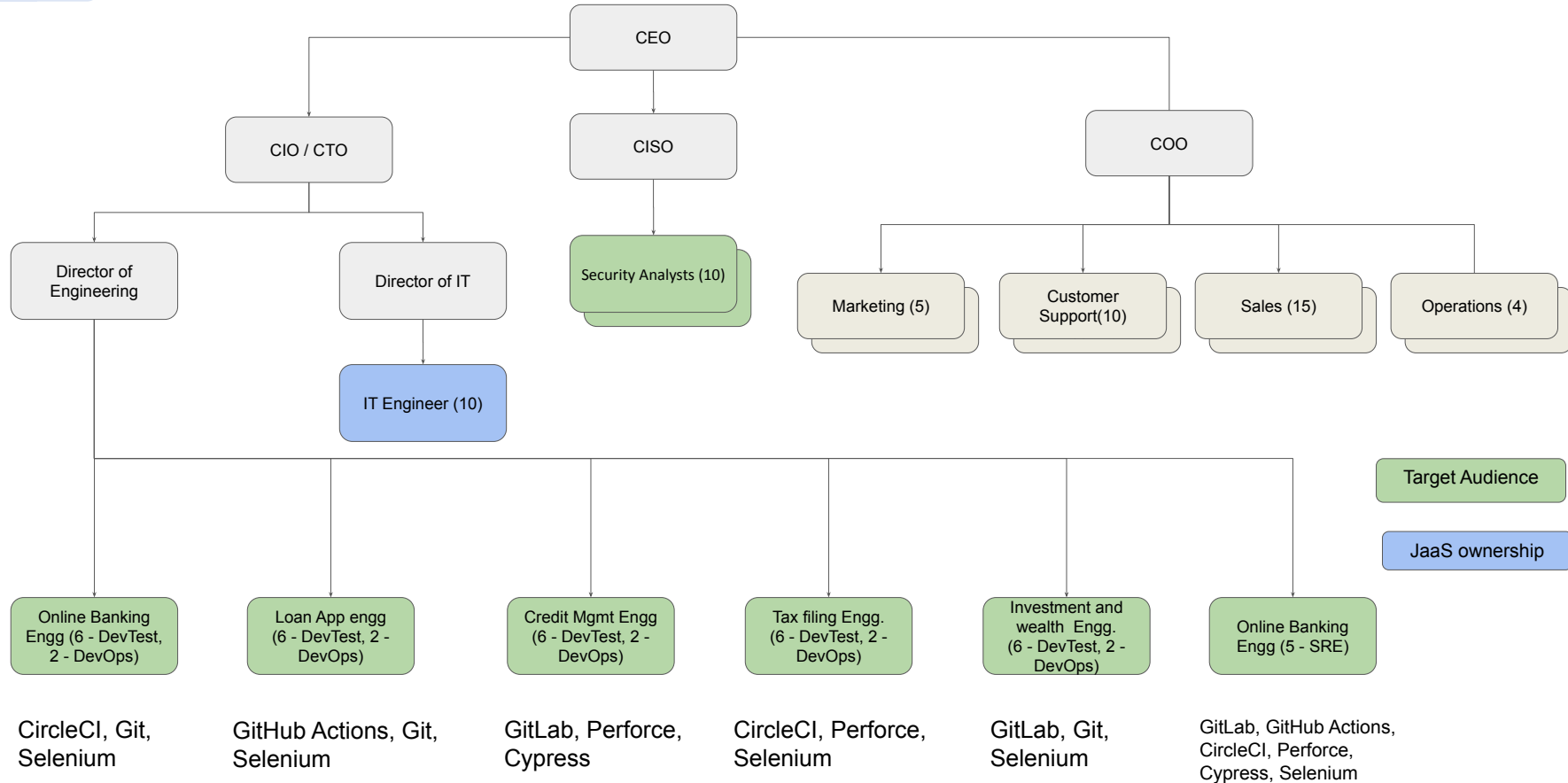
ISync aims to standardize its fragmented CI/CD ecosystem by adopting Jenkins as the unified enterprise solution, which helps us consolidate build and deployment pipelines under a centralized environment.

Employee & Customer growth

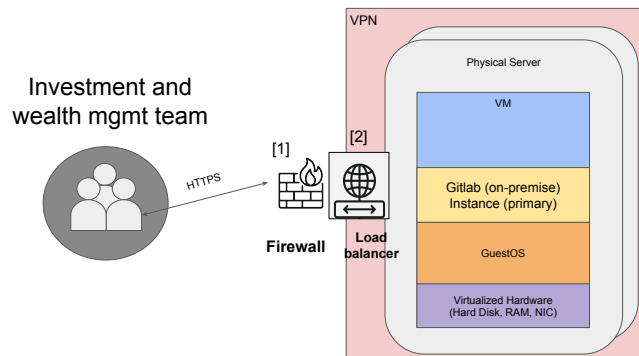
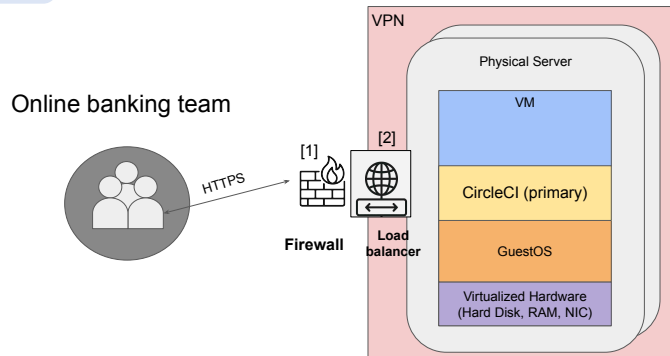


- **Company Name:** FinTech Solutions Inc.
- **Industry:** Financial
- **Company Size:** ~ 100 employees
- **Headquarters:** Santa Clara, California
- **Core Services:** Online banking, Investment and wealth management, tax filing, loan and credit management services
- **Customers' footprint:** United States of America

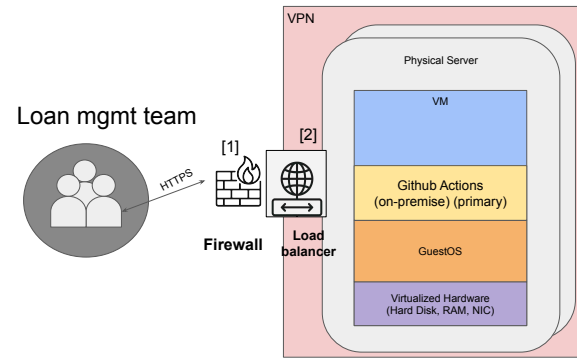
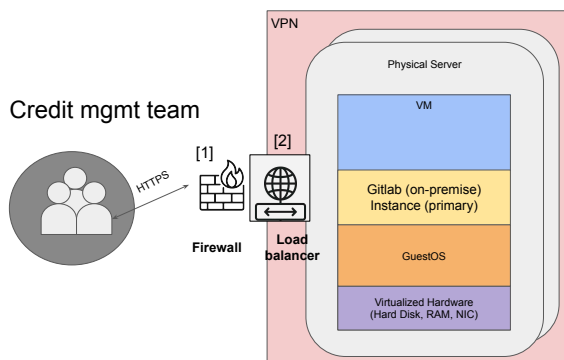
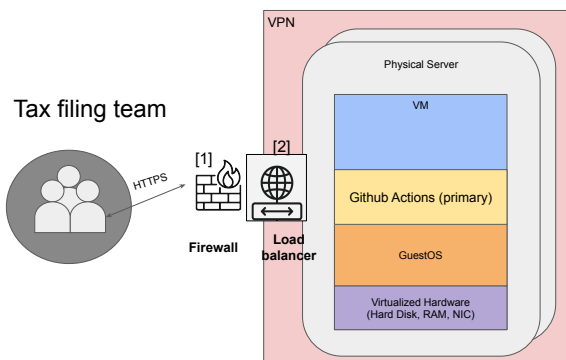
ORGANIZATION STRUCTURE



AS-IS ARCHITECTURE (APP & INFRASTRUCTURE)



Each team runs their own instance of the respective CI/CD tool



[1] firewall image generated with the help of Microsoft Copilot (SCU organization). Refer the attached folder for conversation summary and AI tool confirming the image was generated and not sourced from the web).

[2] load balancer image generated, cropped to fit the requirement for the slide, using Microsoft Copilot (SCU organization) AI tool. Refer the attached folder for conversation summary and AI tool confirming the image was generated and not sourced from the web).

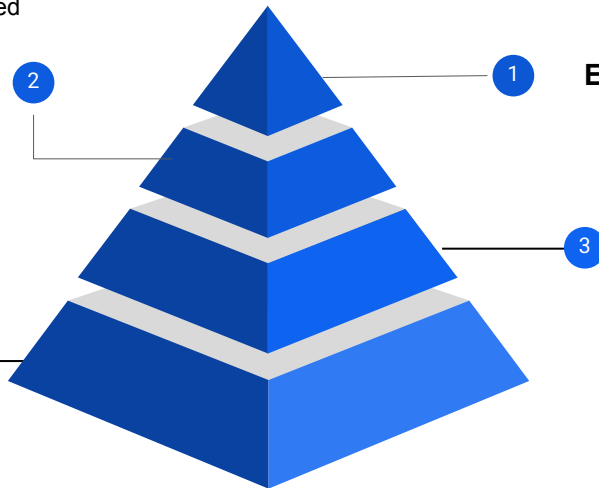
WHY JENKINS^[1]

- With 2000+ plugins^[2] and support for every major build/test tool (Java, Go, Node.js, Python, etc.), Jenkins integrates seamlessly across hybrid and on-prem infrastructures (VMware, Docker, Kubernetes).
- CircleCI, while polished, offers limited extensibility and slower plugin innovation due to its closed ecosystem.

Extensibility and support for DevTools

Cost Efficiency and Vendor Independence

- Jenkins' open-source MIT model removes recurring license fees and vendor lock-in, enabling sustainable CI/CD scaling for ISync.
- CircleCI per-seat pricing and proprietary model can increase costs for large teams.



- Jenkins' self-hosted architecture gives ISync full control over data, security, and compliance—critical for regulated enterprise environments.
- Built on an MIT-licensed open core, ISync can modify, extend, and govern pipelines internally—something CircleCI managed SaaS model restricts.

Enterprise-Grade Control & Governance

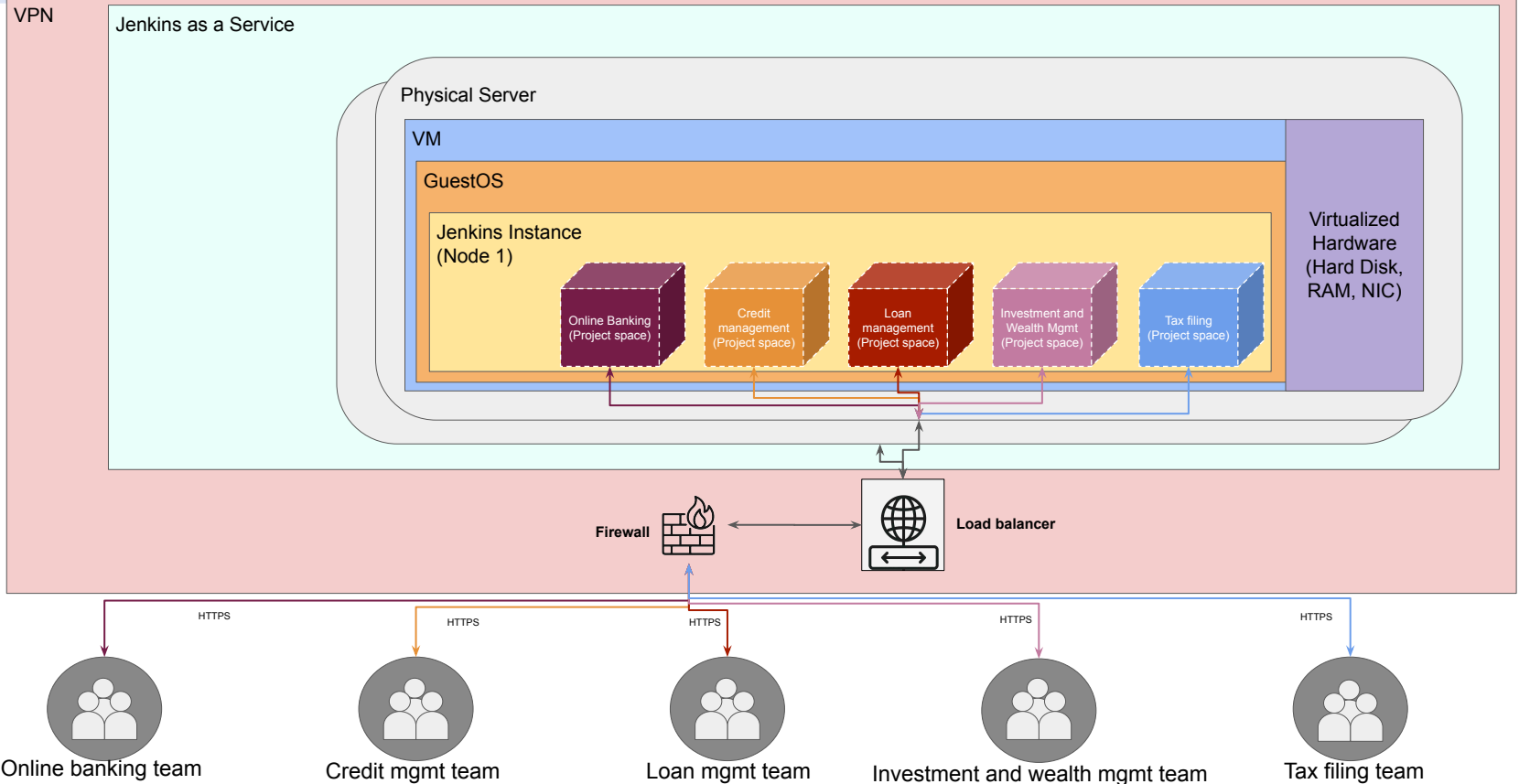
Proven Reliability and Open Innovation

- Backed by the Continuous Delivery Foundation (CDF), Jenkins ensures transparent releases, LTS stability, and long-term community governance.
- CircleCI proprietary updates are managed centrally, giving less visibility and flexibility to enterprise users.

[1] Jenkins contribution reference: <https://www.jenkins.io/blog/2019/05/30/becoming-contributor-newbie-tickets/>

[2] Jenkins plugins marketplace: <https://plugins.jenkins.io/>

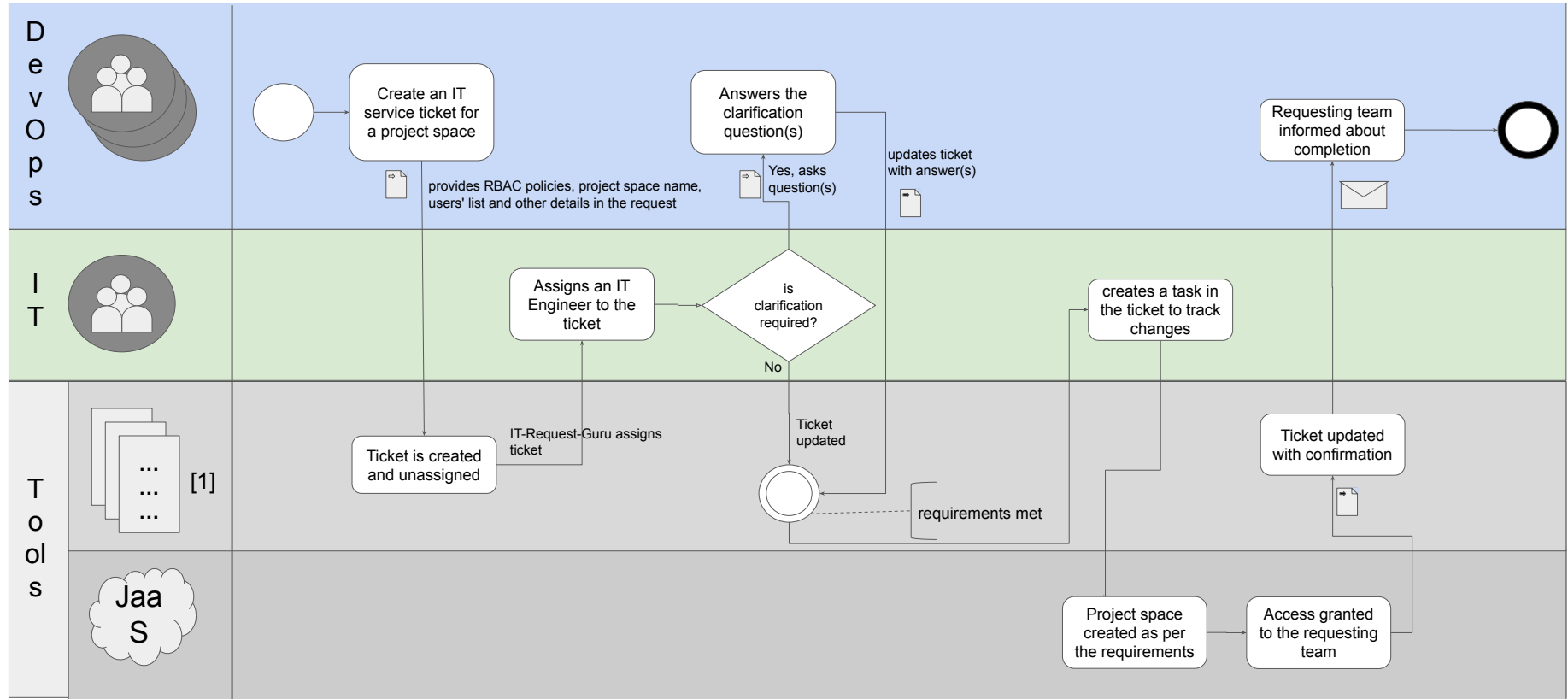
TO-BE ARCHITECTURE (APP & INFRASTRUCTURE)



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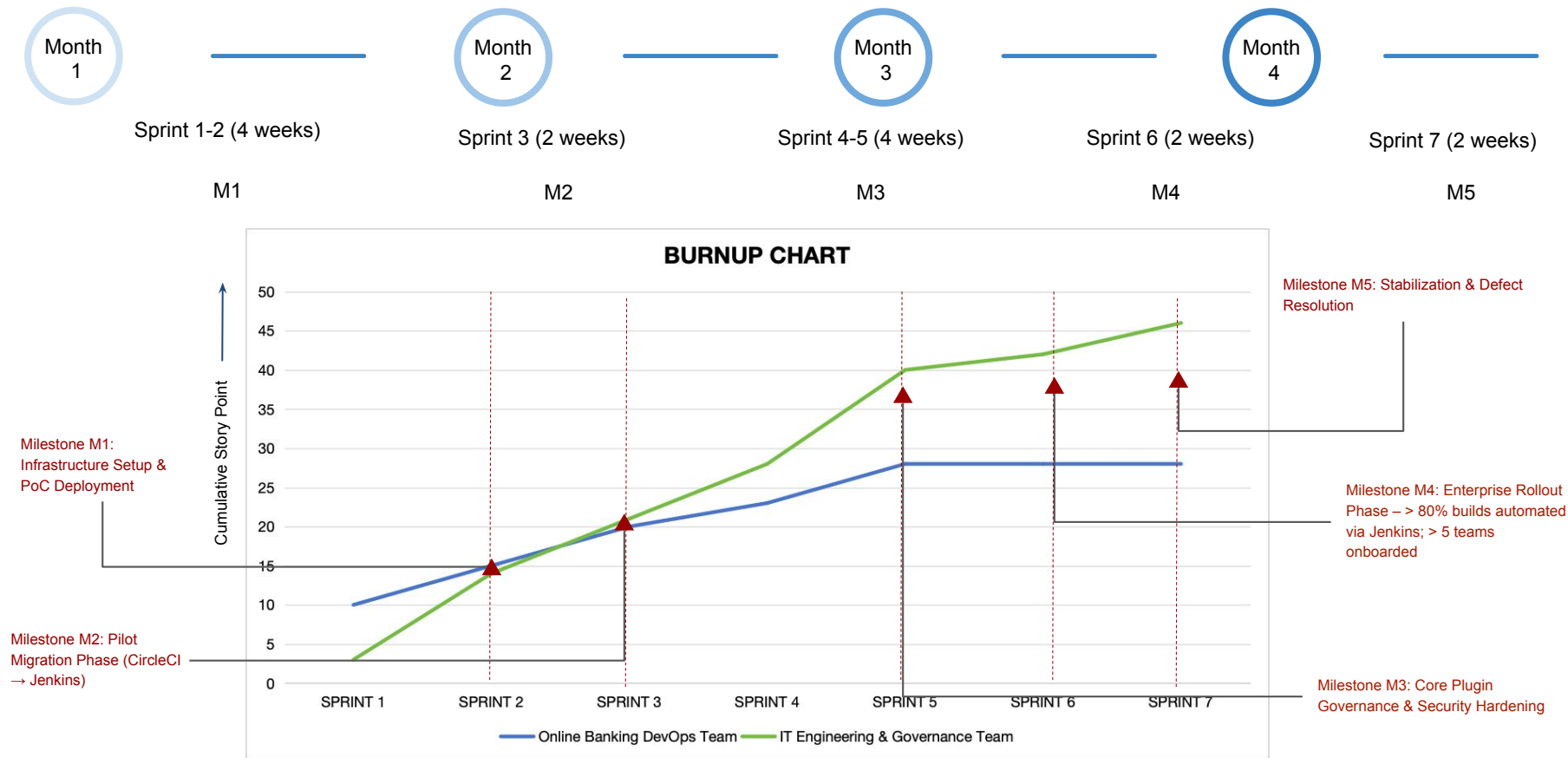
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TO-BE ARCHITECTURE (BPM)



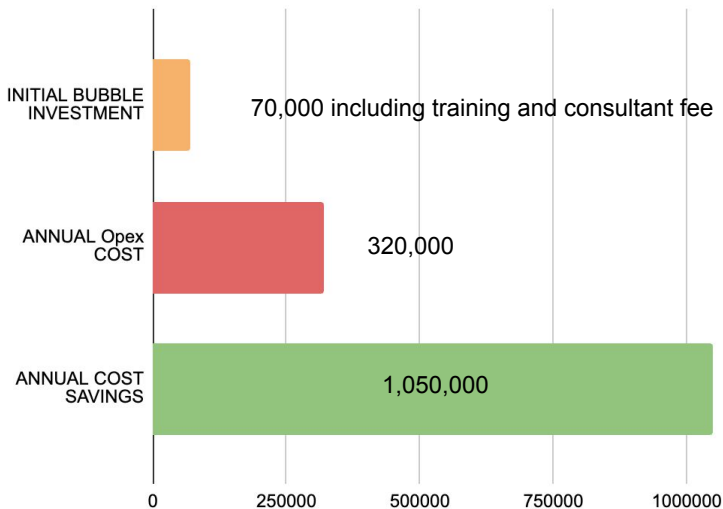
Story ID	Persona	Summary	Value	Time Criticality	Risk Reduction	Job Size (SP)	WSJF Score	Priority Rank
EPIC 1	Shauna	Build CI/CD pipeline on JaaS for online banking	5	5	4	8	1.75	2
US 01.1	Shauna	Pipeline migration from CircleCI to Jenkins	5	4	3	8	1.5	5
US 01.2	Shauna	Jenkins project space + RBAC for banking team	5	5	4	5	2.8	3
US 01.3	Shauna	Enable plugins (Git, JFrog, Selenium)	4	3	3	3	3.3	2
EPIC 2	Joe	Streamline rollback and deployment reliability	4	4	4	8	1.5	5
US 02.1	Joe	Preserve last 5 stable builds (JFrog rollback)	4	4	3	5	2.2	4
US 02.2	Joe	Deploy approved changeset for bug fix	3	5	4	8	1.5	6

BURNUP CHART



ROI & PAYBACK SUMMARY

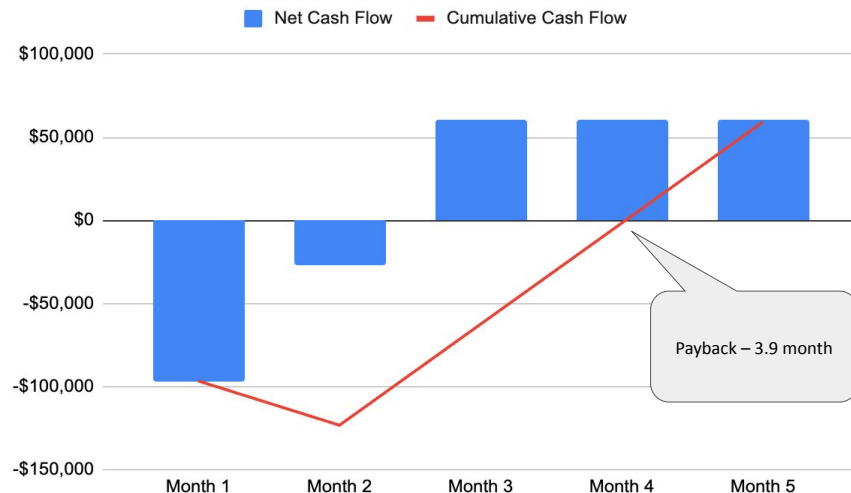
Annual perspective



$$\text{ROI} = \$105,000 - \$70,000 - \$320,000 = \$660,000$$

$$\text{Payback} = (\$320,000 / (\$980,000 / 12)) = \sim 3.9 \text{ month}$$

Monthly perspective



	Benefit	Opex	Net Cash Flow	Cumulative Cash Flow
Month 1	\$0	\$96,667	-\$96,667	-\$96,667
Month 2	\$0	\$26,667	-\$26,667	-\$123,334
Month 3	\$87,500	\$26,667	\$60,833	-\$62,501
Month 4	\$87,500	\$26,667	\$60,833	-\$1,668
Month 5	\$87,500	\$26,667	\$60,833	\$59,165

RISK & MITIGATION

Category	Description of Risk	Potential Impact	Likelihood Level Very low '1' to Very high '5'	Impact Level Very low '1' to very high '5'	Priority Level (Likelihood x Impact)	Mitigation Strategy
Technical Risks[1]	Jenkins or pipeline configuration failures due to misconfigured plugins or updates.	CI/CD processes interrupted; delayed releases and broken builds.	3	4	12	US01.1 - US01.3: Standardize CI/CD pipelines in Jenkins-as-a-Service to remove configuration inconsistencies. Automated setup and plugin control ensure consistent with zero errors deployment across teams.
Operational Risks[2]	VMware or infrastructure downtime causing Jenkins unavailability.	Deployment interruptions; loss of productivity; potential SLA violations.	4	3	12	US02.1 – US02.2: Establish automated rollback pipelines and store the last five stable builds via JFrog. This enables rapid recovery and maintains 99.9% service uptime during failures.
Security & Compliance Risks[3]	Weak RBAC policies or missing audit logs allowing unauthorized access or untracked deployments.	Security breaches, compliance violations, and audit failure during reviews.	2	5	10	US03.1 – US03.2: Enforce RBAC security baselines and a controlled plugin approval workflow to prevent unauthorized access and untracked deployments.
Project Management Risks	Miscommunication between IT and DevOps teams during Jenkins setup or testing.	Schedule delays; unclear ownership; reduced efficiency in coordination.	3	3	9	Hold regular cross-team check-ins; maintain shared documentation; track scope and progress updates through Jira or similar tools.

[1] Cycode published an article about Jenkins configuration and security risks. Retrieved from: <https://cycode.com/blog/jenkins-security-best-practices/>

[2] VMware shared best practices for maintaining high availability and minimizing downtime. Retrieved from: <https://knowledge.broadcom.com/external/article/313917/best-practices-and-advanced-features-for.html>

[3] OWASP Foundation provided guidelines on DevSecOps and RBAC-based security practices. Retrieved from: <https://owasp.org/www-project-devsecops-guideline/>